Factoring Polynomials

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 7$ in solving systems of equations with the elimination method, evaluate $f(5)$.

Final Answer: 27. Substitute $x = 5$: $f(5) = 4(5) + 7 = 27$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 8$ in solving systems of equations with a variety of methods, evaluate $f(2)$.

Final Answer: 18. Substitute $x = 2$: $f(2) = 5(2) + 8 = 18$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for solving systems of equations with a variety of methods consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 21.5. The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 9$ in applications of linear systems, evaluate $f(2)$.

Final Answer: 21. Substitute $x = 2$: $f(2) = 6(2) + 9 = 21$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for applications of linear systems consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 3$ in systems of inequalities, evaluate $f(2)$.

Final Answer: 7. Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for systems of inequalities consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 2 · MediumA sample of a radioactive substance decays such that its mass is halved every 15 days. If the initial mass is 480 grams, which function models the mass $M(t)$, in grams, remaining after t days?

The initial mass is $a = 480$. Halving corresponds to a decay factor of $b = \frac{1}{2}$. Since halving occurs every 15 days, the exponent must count how many 15-day intervals have elapsed, which is $\frac{t}{15}$: $M(t) = 480\left(\frac{1}{2}\right)^{\frac{t}{15}}$.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Level 4 · SynthesisA radioactive isotope decays by 20% every 15 years. If the initial mass is 500 grams, which expression represents the remaining mass, in grams, after m months?

First convert 15 years into months: $15 \times 12 = 180$ months per decay period. Since the mass decreases by 20% each period, the retention factor is $1 - 0.20 = 0.80$. After m months, the number of 180-month periods is $\frac{m}{180}$: $M(m) = 500(0.80)^{m/180}$.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 1 · FoundationsWhich expression is equivalent to $\frac{x^{5/3}}{x^{2/3}} \cdot \sqrt{x}$ for all positive values of x?

Step 1: Subtract powers for the quotient: $\frac{x^{5/3}}{x^{2/3}} = x^{5/3\,-\,2/3} = x^{3/3} = x^1$. Step 2: Multiply by $\sqrt{x} = x^{1/2}$: $x^1 \cdot x^{1/2} = x^{1\,+\,1/2} = x^{3/2} = \sqrt{x^3}$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Use the quadratic formula to solve $x^2 + 2x - 8 = 0$.

Step 1: Identify coefficients: $a = 1$, $b = 2$, $c = -8$. Step 2: Apply the quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{2^2 - 4(1)(-8)}}{2(1)} = \frac{-2 \pm \sqrt{4 + 32}}{2} = \frac{-2 \pm \sqrt{36}}{2}$ Step 3: Evaluate roots: $x = \frac{-2 \pm 6}{2} \Rightarrow x_1 = \frac{4}{2} = \mathbf{2}, \quad x_2 = \frac{-8}{2} = \mathbf{-4}$ The solutions are $\mathbf{x = 2}$ and $\mathbf{x = -4}$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A solid toy is formed by attaching a cone of radius 4 cm and height 9 cm to the flat face of a solid hemisphere of radius 4 cm. Calculate the total volume of the toy. Give your answer correct to 3 significant figures.

Step 1: Volume of cone: $V_{\text{cone}} = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(4^2)(9) = 48\pi$ cm^3 Step 2: Volume of hemisphere: $V_{\text{hemi}} = \frac{1}{2} \times \frac{4}{3}\pi r^3 = \frac{2}{3}\pi(4^3) = \frac{128}{3}\pi$ cm^3 Step 3: Total volume: $V_{\text{total}} = 48\pi + \frac{128}{3}\pi = \frac{272}{3}\pi \approx 284.838\ldots \approx 285$ cm^3 (3 s. f.)

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A school theater sold 250 tickets for a play, raising a total of $1,750. Adult tickets cost $9 each and student tickets cost $5 each. How many adult tickets were sold?

Set up system of equations: Let a be the number of adult tickets and s be student tickets: $\{a + s = 250; 9a + 5s = 1750$ Elimination method: Multiply the first equation by 5: $5a + 5s = 1250$. Subtract from the second equation: $(9a + 5s) - (5a + 5s) = 1750 - 1250 \Rightarrow 4a = 500 \Rightarrow \mathbf{a = 125}$. Exactly 125 adult tickets were sold (and $s = 250 - 125 = 125$ student tickets).

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a right triangle, acute angle θ satisfies $\sin\theta = \frac{5}{13}$. Find the exact values of $\cos\theta$, $\tan\theta$, and $\sec\theta$.

Step 1: In a right triangle with opposite side 5 and hypotenuse 13, the adjacent side is: $\text{adj} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$. Step 2: Compute trigonometric ratios:$\cos\theta = \frac{\text{adj}}{\text{hyp}} = \mathbf{\frac{12}{13}}$. $\tan\theta = \frac{\text{opp}}{\text{adj}} = \mathbf{\frac{5}{12}}$.$\sec\theta = \frac{1}{\cos\theta} = \mathbf{\frac{13}{12}}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Quadratic Functions

Write the general circle equation $x^2 + y^2 - 8x + 6y + 9 = 0$ in standard form, identify its center and radius, and find the equation of the tangent line to the circle at the point $P(4, 1)$.

Complete the square for x and y: $(x^2 - 8x + 16) + (y^2 + 6y + 9) = -9 + 16 + 9 = 16$. $(x - 4)^2 + (y + 3)^2 = 16$. Center is $C(4, -3)$ and radius is $r = \sqrt{16} = 4$. Check the point $P(4, 1)$: $(4 - 4)^2 + (1 + 3)^2 = 0 + 16 = 16$ ✓. The radius connects $C(4, -3)$ to $P(4, 1)$, which is a vertical line $x = 4$. Therefore, the tangent line through $P(4, 1)$ is horizontal: $y = 1$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Exponents & Polynomials

Determine the equation of the circle in the xy-plane for which the segment connecting $A(2, -3)$ and $B(8, 5)$ serves as a diameter.

Step 1: The center (h, k) is the midpoint of diameter AB: $(h, k) = \left(\frac{2+8}{2}, \frac{-3+5}{2}\right) = \left(\frac{10}{2}, \frac{2}{2}\right) = \mathbf{(5, 1)}$. Step 2: Compute radius r as the distance from center $(5, 1)$ to $B(8, 5)$: $r^2 = (8 - 5)^2 + (5 - 1)^2 = 3^2 + 4^2 = 9 + 16 = 25 \Rightarrow r = 5$. Step 3: Write standard equation: $\mathbf{(x - 5)^2 + (y - 1)^2 = 25}$.

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17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 4.01 Functions & Transformations**

A fair die is rolled twice. Find $P(\text{both show } 6)$.

Step 1: Determine independent probabilities: For a fair six-sided die, the probability of rolling a 6 on any single roll is $P(6) = \frac{1}{6}$. Step 2: Apply the multiplication rule for independent events: $P(\text{both } 6) = P(6) \times P(6) = \frac{1}{6} \times \frac{1}{6} = \mathbf{\frac{1}{36}} \approx 0.0278$ The probability is $\mathbf{\frac{1}{36}}$ (or 1 out of 36 outcomes).

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 5.01 Exponential & Logarithmic Functions**

Suppose y varies directly as x^2 and inversely as z. If $y = 18$ when $x = 3$ and $z = 4$, find the value of y when $x = 5$ and $z = 10$.

Step 1: Formulate the joint variation equation: $y = k\frac{x^2}{z}$ Step 2: Solve for constant k using the given initial condition: $18 = k\frac{3^2}{4} = k\frac{9}{4} \Rightarrow k = 18 \times \frac{4}{9} = 2 \times 4 = 8$. Step 3: Calculate y for $x = 5$ and $z = 10$: $y = 8\frac{5^2}{10} = 8\frac{25}{10} = 8 \times 2.5 = \mathbf{20}$. The value of y is $\mathbf{20}$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 6.01 Systems & Inequalities**

A car travels at 60 mph for 2.5 hours. Find the distance.

Apply the uniform distance formula $d = r \cdot t$: $d = 60 \text{ miles/hour} \times 2.5 \text{ hours} = 60 \times \frac{5}{2} = \mathbf{150}$ miles Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 7.01 Sequences & Series**

A jacket costs USD 50 after a 20% discount. Find the original price.

Let P be the original price. A 20% discount leaves $100\% - 20\% = 80\%$ of the original price: $0.80 \times P = 50$ $P = \frac{50}{0.80} = \frac{50}{4/5} = 50 \times \frac{5}{4} = \frac{250}{4} = \mathbf{USD62.50}$ Trap Warning: Do NOT take 20% of $USD50$ and add it ($USD50 + USD10 = USD60$); percentages must always be calculated relative to the original base!

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 5$ in graphing systems of equations, evaluate $f(2)$.

a. 9 ✓

b. 11

c. 4

d. 13

Substitute $x = 2$: $f(2) = 2(2) + 5 = 9$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 6$ in solving systems of equations with the substitution method, evaluate $f(2)$.

a. 12 ✓

b. 15

c. 6

d. 16

Substitute $x = 2$: $f(2) = 3(2) + 6 = 12$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for graphing systems of equations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solving systems of equations with the substitution method consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the quadratic function $y = ax^2 + bx + c$ is shown below. The parabola intersects the x-axis at points $(1, 0)$ and $(p, 0)$, and has a y-intercept of $(0, -3)$. If the axis of symmetry of the parabola is the line $x = 2$, find the value of p.

a. **3 ✓**

b. 4

c. 5

d. 6

The roots of the quadratic equation $ax^2 + bx + c = 0$ are the x-intercepts of the parabola. We are given one root as $x_1 = 1$. The axis of symmetry of a parabola given by $y = ax^2 + bx + c$ is $x = -\frac{b}{2a}$. We are given that the axis of symmetry is $x = 2$. For a quadratic function, the axis of symmetry is located exactly halfway between the roots. If the roots are x_1 and x_2, then the axis of symmetry is $x = \frac{x_1 + x_2}{2}$. We are given $x_1 = 1$ and the axis of symmetry $x = 2$. Let the oth.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of the quadratic function $y = x^2 - 6x + 8$ is shown below. What are the coordinates of the vertex of the parabola?

a. **(3, -1) ✓**

b. (3, 1)

c. (1, 3)

d. (-1, 3)

The vertex form of a quadratic equation is $y = a(x - h)^2 + k$, where (h, k) is the vertex. We can convert the given equation $y = x^2 - 6x + 8$ to vertex form by completing the square. First, group the x terms: $y = (x^2 - 6x) + 8$. To complete the square for $x^2 - 6x$, take half of the coefficient of x (-6), square it ((-3)^2 = 9), and add and subtract it inside the parentheses: $y = (x^2 - 6x + 9 - 9) + 8$. Rearrange to form the perfect square trinomial: $y = (x^2 - 6x + 9) - 9 + 8$. Simplify.

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the polynomial function $P(x) = x^3 - 6x^2 + 11x - 6$. If $P(x)$ can be factored into the form $(x - a)(x - b)(x - c)$, where $a, b,$ and c are distinct real numbers, what is the value of $a^2 + b^2 + c^2$?

a. **14** ✓

b. 29

c. 38

d. 41

We are given the polynomial $P(x) = x^3 - 6x^2 + 11x - 6$. If $P(x)$ can be factored into $(x - a)(x - b)(x - c)$, then by Vieta's formulas, the sum of the roots is $a + b + c = -(-6)/1 = 6$, the sum of the products of the roots taken two at a time is $ab + ac + bc = 11/1 = 11$, and the product of the roots is $abc = -(-6)/1 = 6$. We need to find $a^2 + b^2 + c^2$. We know the algebraic identity $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + ac + bc)$. Rearranging this identity, we get $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + ac + bc)$. Substitutin.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A biologist is modeling the population of a rare plant species. The population, $P(t)$, in thousands, after t years is approximated by the function $P(t) = a \cdot 2^{kt}$, where a and k are constants. If the initial population in year $t = 0$ was 10,000 plants and after 3 years the population had grown to 80,000 plants, what is the approximate population after 6 years, in thousands?

a. 320

b. 640

c. **1280** ✓

d. 2560

The population model is given by $P(t) = a \cdot 2^{kt}$. We are given that at $t = 0$, the population is 10,000, so $P(0) = 10$. Substituting this into the model: $10 = a \cdot 2^{k \cdot 0}$ $10 = a \cdot 2^0$ $10 = a \cdot 1$ $a = 10$. So, the population model is $P(t) = 10 \cdot 2^{kt}$. We are also given that at $t = 3$, the population is 80,000, so $P(3) = 80$. Substituting this into the model: $80 = 10 \cdot 2^{k \cdot 3}$ Divide both sides by 10: $8 = 2^{3k}$ Since $8 = 2^3$, we have: $2^3 = 2^{3k}$ Equating th.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A set of exam scores is normally distributed with a mean of $\mu = 74$ and a standard deviation of $\sigma = 8$. What is the z-score corresponding to an exam score of 58?

a. -2 ✓

b. -0.5

c. 0.5

d. 2

To find the z-score corresponding to a given data value, use the standard normal distribution formula: $z = \frac{x - \mu}{\sigma}$ where x is the individual score, μ is the population mean, and σ is the standard deviation. Substitute the given values into the formula: $z = \frac{58 - 74}{8} = \frac{-16}{8} = -2$ A z-score of -2 indicates that the score of 58 is located 2 standard deviations below the mean. Common Misconceptions: - 2: Results from subtracting the score from the mean.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A farmer wants to fence a rectangular field adjacent to a straight river. The farmer has 120 meters of fencing material. The side of the field parallel to the river does not require fencing. Let the dimensions of the field perpendicular to the river be x meters, and the dimension parallel to the river be y meters. The area of the field is given by $A(x) = 120x - 2x^2$. What is the maximum possible area of the field?

a. **1800 square meters** ✓

b. 2000 square meters

c. 2400 square meters

d. 2800 square meters

The area of the rectangular field is given by the quadratic function $A(x) = 120x - 2x^2$. This is a downward-opening parabola, and its maximum value occurs at the vertex. The x-coordinate of the vertex of a parabola in the form $ax^2 + bx + c$ is given by $-b/(2a)$. In this case, $a = -2$ and $b = 120$. The x-coordinate of the vertex is $x = -120/(2 \times -2) = -120/-4 = 30$. This value of x represents the dimension perpendicular to the river that maximizes the area. To find the maximum area, substitute.

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31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The graph of the function $f(x) = ax^2 - 8x + c$ has its vertex at $(2, -4)$. What is the value of $a + c$?

a. -6

b. -5

c. **5** ✓

d. 6

The vertex of a parabola given by $f(x) = Ax^2 + Bx + C$ is at $x = -\frac{B}{2A}$. In this case, $A = a$, $B = -8$, and $C = c$. The x-coordinate of the vertex is given as 2. Thus, $2 = -\frac{-8}{2a} = \frac{8}{2a} = \frac{4}{a}$. Solving for a, we get $a = \frac{4}{2} = 2$. Now that we know $a = 2$, the function is $f(x) = 2x^2 - 8x + c$. The vertex is at $(2, -4)$, which means $f(2) = -4$. Substituting $x = 2$ into the function: $f(2) = 2(2)^2 - 8(2) + c$ $-4 = 2(4) - 16 + c$ $-4 = 8 - 16 + c$ $-4 = -8 + c$ Solving f.

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The parabola has its vertex at $(-2, 8)$ and passes through the point $(0, 4)$. Determine the value of c.

a. 4 ✓

b. -2

c. 8

d. 2

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. The graph of this function is a parabola. The vertex of the parabola is given as $(-2, 8)$. A key property of a parabola is that the y-intercept occurs where $x = 0$. We are given that the parabola passes through the point $(0, 4)$. This means that when $x = 0$, $f(x) = 4$. Substituting $x = 0$ into the general form of the quadratic function, we get: $f(0) = a(0)^2 + b(0) + c$ $f(0) = 0 + 0 + c$ $f(0) = c$ Since we know that $f(0) = 4$, we can conclude t.

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The graph of the function $y = \sqrt{x - a} + b$ is shown below. If the vertex of the graph is at $(7, 3)$, what is the value of $a + b$?

a. 3

b. 4

c. **10** ✓

d. 11

The graph of the function $y = \sqrt{x - a} + b$ is a square root function. The vertex of such a function occurs at the point where the expression under the square root is zero, and the square root itself is zero. Thus, the vertex occurs at $x - a = 0$ and $y = 0 + b$. This means the vertex is at (a, b). We are given that the vertex is at $(7, 3)$. Therefore, we can equate the coordinates: $a = 7$ and $b = 3$. The question asks for the value of $a + b$. $a + b = 7 + 3 = 10$.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the sum of the roots of the quadratic equation $3x^2 - 11x + 2 = 0$?

a. $-\frac{2}{3}$

b. $-\frac{11}{3}$

c. $\frac{11}{3}$ ✓

d. $\frac{2}{3}$

For any quadratic equation $ax^2 + bx + c = 0$, Vieta's formulas state that: - Sum of roots: $r_1 + r_2 = -\frac{b}{a}$ - Product of roots: $r_1 r_2 = \frac{c}{a}$ Here, $a = 3$, $b = -11$, and $c = 2$. Thus, the sum of the roots is: $r_1 + r_2 = -\frac{-11}{3} = \frac{11}{3}$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A psychologist conducted an experiment where 80 participants were randomly assigned to either meditate for 15 minutes before an exam or sit quietly. The meditation group scored significantly higher on average. To what population can the causal conclusion be safely generalized?

a. All college students worldwide

b. All adults of any age

c. **Individuals similar to the participants in the study** ✓

d. No causal conclusions can be made

Because treatments were randomly assigned, causality is established. However, because the participants were not randomly selected from the entire world population, the causal finding can only be generalized to individuals similar to the participants in the study.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $\sqrt{2x + 15} - x = 0$, what is the sum of all valid real solutions?

a. 3

b. **5** ✓

c. 7

d. 8

$\sqrt{2x + 15} = x$. Square both sides: $2x + 15 = x^2 \Rightarrow x^2 - 2x - 15 = 0 \Rightarrow (x - 5)(x + 3) = 0$. At $x = -3$: $\sqrt{9} = 3eq - 3$ (extraneous). At $x = 5$: $\sqrt{25} = 5$ (valid). The only valid solution is 5.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the value of x if $\frac{2^x}{2^3} = 16$?

a. 4

b. **5** ✓

c. 7

d. 19

Using the property of exponents $\frac{a^m}{a^n} = a^{m-n}$, the equation can be rewritten as $2^{x-3} = 16$. Since $16 = 2^4$, we have $2^{x-3} = 2^4$. Equating the exponents, we get $x - 3 = 4$. Adding 3 to both sides gives $x = 7$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

What is the extraneous solution to the radical equation $\sqrt{x+18} = x - 2$?

a. **-2** ✓

b. -3

c. 5

d. 7

Square both sides: $x + 18 = x^2 - 4x + 4 \Rightarrow x^2 - 5x - 14 = 0 \Rightarrow (x-7)(x+2) = 0$. At $x = 7$, $\sqrt{25} = 5 = 7 - 2$ (valid). At $x = -2$, $\sqrt{16} = 4eq - 4$ (extraneous). The extraneous solution is -2.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Systems & Inequalities

For what positive value of c does the equation $2x^2 - 4x + c = 0$ have exactly one distinct real solution?

a. 16

b. 6

c. 2 ✓

d. 4

A quadratic equation $ax^2 + bx + c = 0$ has exactly one distinct real solution (a double root) when its discriminant $\Delta = b^2 - 4ac$ is equal to zero: $(-4)^2 - 4(2)(c) = 0$ $16 - 8c = 0$ $8c = 16$ $c = \frac{16}{8} = 2$.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Sequences & Series

What are all real solutions to the rational equation $\frac{x}{x-3} = \frac{3}{x-3} + 4$?

a. $x = 3$

b. $x = 4$

c. $x = 7$

d. **No real solution ✓**

Multiply by $x - 3$: $x = 3 + 4(x - 3) \Rightarrow x = 4x - 9 \Rightarrow 3x = 9 \Rightarrow x = 3$. However, substituting $x = 3$ makes the denominator zero. Therefore $x = 3$ is extraneous and there is no real solution.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Radicals & Exponents

If $(x - 3)$ is a factor of the polynomial $P(x) = 2x^3 - kx^2 + 1x - 3$, what is the value of the constant k?

a. 6 ✓

b. 3

c. 8

d. -6

By the Factor Theorem, if $(x - 3)$ is a factor of $P(x)$, then $P(3) = 0$. Substitute $x = 3$ into $P(x)$: $P(3) = 2(3)^3 - k(3)^2 + 1(3) - 3 = 0\ 54 - 9k + 3 - 3 = 0\ 54 = 9k$ Divide by 9: $k = 6$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Linear Equations & Lines

What is the valid real solution to the equation $\frac{x}{x-2} + \frac{1}{x+2} = \frac{8}{x^2-4}$?

a. **-5 ✓**

b. -2

c. 2

d. 5

Multiply by $x^2 - 4$: $x(x + 2) + (x - 2) = 8 \Rightarrow x^2 + 3x - 10 = 0 \Rightarrow (x + 5)(x - 2) = 0$. At $x = 2$, the denominator is zero (extraneous). The only valid solution is $x = -5$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Rational Expressions

If $x = 3$ is a solution to the quadratic equation $2x^2 + kx - 15 = 0$, what is the value of the constant k?

a. -1 ✓

b. 1

c. -3

d. 3

Substitute $x = 3$ into the equation: $2(3)^2 + k(3) - 15 = 0 \Rightarrow 2(9) + 3k - 15 = 0 \Rightarrow 18 - 15 + 3k = 0 \Rightarrow 3 + 3k = 0 \Rightarrow 3k = -3 \Rightarrow k = -1$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Statistics & Data Analysis

Consider the quadratic function $f(x) = x^2 - 6x + 9$. What is the value of $f(3)$?

a. 0 ✓

b. 3

c. 9

d. 18

To find the value of $f(3)$, we substitute $x = 3$ into the function $f(x) = x^2 - 6x + 9$. $f(3) = (3)^2 - 6(3) + 9$ $f(3) = 9 - 18 + 9$ $f(3) = 0$ Therefore, the value of $f(3)$ is 0.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Quadratic Functions

Section 1 (calculator-allowed) has 24 questions completed in 30 minutes; Section 2 has 20 questions completed in 20 minutes. Keeping each section's original per-question pace, how many total minutes would be needed if each section instead had 30 questions?

a. 62.5 minutes

b. 50.0 minutes

c. 57.5 minutes

d. **67.5 minutes** ✓

Section 1's pace is $30/24 = 1.25$ min/question, giving $30(1.25) = 37.5$ minutes; Section 2's pace is $20/20 = 1.0$ min/question, giving 30 minutes. Total: $37.5 + 30 = 67.5$ minutes.

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